Task 7 Interview Questions and Model Answers for Data Analyst Internship

How did you connect Python to a database?

I used Python's built-in sqlite3 library to establish a connection to the SQLite database file. For example:

```
import sqlite3
conn = sqlite3.connect("sales_data.db")
```

This creates a Connection object `conn` that allows executing SQL queries against the `sales_data.db` file.

What SQL query did you run?

I ran a query to summarize sales by product, calculating total quantity and total revenue:

```
SELECT
    product,
    SUM(quantity) AS total_qty,
    SUM(quantity * price) AS revenue
FROM sales
GROUP BY product;
```

This groups rows by product and aggregates the quantity and revenue per product.

What does GROUP BY do?

The `GROUP BY` clause groups records that have the same values in the specified column(s). It allows aggregate functions like `SUM()`, `AVG()`, or `COUNT()` to produce summarized results per group rather than across the entire table.

How did you calculate revenue?

Revenue was calculated in the SQL query by multiplying the `quantity` by the `price` for each row and then summing those products per product:

```
SUM(quantity * price) AS revenue
```

This returns the total sales amount per product.

How did you visualize the result?

I loaded the SQL result into a pandas DataFrame and then used matplotlib via pandas plotting:

```python
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_sql_query(query, conn)
df.plot(kind='bar', x='product', y='revenue')
plt.title('Revenue by Product')
plt.xlabel('Product')
plt.ylabel('Revenue')
plt.savefig('sales_chart.png')
plt.show()
```

This produces a bar chart showing revenue per product.

What does pandas do in your code?

Pandas provides a convenient `read_sql_query` function to run SQL queries directly into a DataFrame. It then offers easy data manipulation and built-in plotting support that integrates with matplotlib, simplifying data analysis workflows.

What's the benefit of using SQL inside Python?

Using SQL inside Python combines the power of SQL for efficient data retrieval and aggregation with Python's extensive libraries (like pandas and matplotlib) for further processing, analysis, and visualization within a single environment. It streamlines workflows and reduces context switching.

Could you run the same SQL query directly in DB Browser for SQLite?

Yes. The same `SELECT` query can be executed in DB Browser for SQLite's SQL tab. It would produce identical aggregated results, but without Python's downstream data manipulation and visualization capabilities.